

-- SECTION 1

CREATE DATABASE IF NOT EXISTS hive_test;

USE hive_test;

-- SALESMAN

```
CREATE TABLE salesman (  
    salesman_id INT,  
    name STRING,  
    city STRING,  
    commission DOUBLE  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
STORED AS TEXTFILE;
```

```
LOAD DATA LOCAL INPATH '/path/to/salesman.csv'  
INTO TABLE salesman;
```

-- CUSTOMER

```
CREATE TABLE customer (  
    customer_id INT,  
    cust_name STRING,  
    city STRING,  
    grade INT,  
    salesman_id INT  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
STORED AS TEXTFILE;
```

```
LOAD DATA LOCAL INPATH '/path/to/customer.csv'  
INTO TABLE customer;
```

-- ORDERS

```
CREATE TABLE orders (  
    ord_no INT,  
    purch_amt DOUBLE,
```

```
    ord_date DATE,  
    customer_id INT,  
    salesman_id INT  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
STORED AS TEXTFILE;
```

```
LOAD DATA LOCAL INPATH '/path/to/orders.csv'  
INTO TABLE orders;
```

```
-- =====  
-- FIVE SQL QUESTIONS  
-- =====
```

```
-- 1. Same city  
SELECT  
    s.name AS salesman_name,  
    c.cust_name AS customer_name,  
    s.city  
FROM salesman s  
JOIN customer c  
    ON s.city = c.city;
```

```
-- 2. Salesman and customer  
SELECT  
    s.name AS salesman_name,  
    c.cust_name AS customer_name  
FROM salesman s  
JOIN customer c  
    ON s.salesman_id = c.salesman_id;
```

```
-- 3. Orders between 500 and 2000  
SELECT  
    o.ord_no,  
    o.purch_amt,  
    c.cust_name,  
    c.city  
FROM orders o  
JOIN customer c
```

```
    ON o.customer_id = c.customer_id
WHERE o.purch_amt BETWEEN 500 AND 2000;
```

```
-- 4. Commission greater than 12%
```

```
SELECT
    c.cust_name,
    s.name AS salesman_name,
    s.commission
FROM customer c
JOIN salesman s
    ON c.salesman_id = s.salesman_id
WHERE s.commission > 0.12;
```

```
-- 5. Different city + commission > 12%
```

```
SELECT
    c.cust_name,
    c.city AS customer_city,
    s.name AS salesman_name,
    s.city AS salesman_city,
    s.commission
FROM customer c
JOIN salesman s
    ON c.salesman_id = s.salesman_id
WHERE c.city <> s.city
    AND s.commission > 0.12;
```

```
-- =====
-- SECTION 2
-- COMPLEX TABLE
-- =====
```

```
CREATE TABLE salesdetail_complex (
    Product_ID INT,
    productdetails MAP<STRING,STRING>,
    Order_Priority VARCHAR(4),
    merchantType CHAR(4),
    Sale_Amount DOUBLE,
    Order_Quantity BIGINT,
    Discount FLOAT,
    Salaryhike TINYINT,
    companyprofit SMALLINT,
```

```

    financeDeficit DECIMAL(8,2),
    indian BOOLEAN,
    saledate ARRAY<DATE>,
    saleyear ARRAY<INT>,
    selleramountfile ARRAY<DOUBLE>,
    orderQuantityfile ARRAY<BIGINT>,
    costlist MAP<INT,INT>,
    strutureType STRUCT<
        city:STRING,
        state:STRING,
        pin:BIGINT
    >,
    systemdatetime ARRAY<STRING>
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
COLLECTION ITEMS TERMINATED BY '$'
MAP KEYS TERMINATED BY '#'
STORED AS TEXTFILE;

-- Load complex data
LOAD DATA LOCAL INPATH '/path/to/salesdetail_complex.csv'
INTO TABLE salesdetail_complex;

-- Get only 2 records
SELECT *
FROM salesdetail_complex
LIMIT 2;

-- =====
-- NON-PARTITIONED TABLE
-- =====

CREATE TABLE non_part (
    dateid SMALLINT,
    caldate DATE,
    day STRING,
    week SMALLINT,
    month STRING,
    qtr STRING,
    year SMALLINT,

```

```
    holiday BOOLEAN
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY '|'
STORED AS TEXTFILE;
```

```
LOAD DATA LOCAL INPATH '/path/to/non_part.csv'
INTO TABLE non_part;
```

```
-- =====
-- PARTITIONED TABLE
-- =====
```

```
CREATE TABLE date_part (
    dateid SMALLINT,
    day STRING,
    week SMALLINT,
    month STRING,
    qtr STRING,
    year SMALLINT,
    holiday BOOLEAN
)
PARTITIONED BY (
    caldate DATE
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY '|'
STORED AS TEXTFILE;
```

```
INSERT OVERWRITE TABLE date_part
PARTITION (caldate)
SELECT
    dateid,
    day,
    week,
    month,
    qtr,
    year,
    holiday,
    caldate
FROM non_part;
```

```
-- =====  
-- PARTITIONED + BUCKETED TABLE  
-- =====
```

```
CREATE TABLE date_part_bucket (  
    dateid SMALLINT,  
    week SMALLINT,  
    month STRING,  
    qtr STRING,  
    year SMALLINT,  
    holiday BOOLEAN  
)  
PARTITIONED BY (  
    caldate DATE  
)  
CLUSTERED BY (dateid)  
INTO 4 BUCKETS  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY '|'   
STORED AS TEXTFILE;
```

```
SET hive.enforce.bucketing = true;
```

```
INSERT OVERWRITE TABLE date_part_bucket  
PARTITION (caldate)  
SELECT  
    dateid,  
    week,  
    month,  
    qtr,  
    year,  
    holiday,  
    caldate  
FROM non_part;
```